

Output Ontology (OO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: ARABICMMLU | Context: MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The flat single-correct-answer MCQ schema with 2–5 options [Q27] structurally matches the deployment's MCQ tutoring use case, and the user explicitly noted that for MCQ tasks explanation format is not exam-critical [elicitation A3]. This is a meaningful structural alignment. However, the schema provides no mechanism for jurisdiction-specific or country-specific correct answers. The deployment user explicitly stated 'the system must tailor legal answers to each country's specific legal system' [elicitation A2], but the benchmark's flat label structure encodes a single ground-truth answer per question [Q27], with no country-stratified branching. The dataset analysis empirically confirms this: a Moroccan procedural-law answer is simply marked correct without any 'correct-for-Morocco' qualifier [DATASET-D8, D9, D10]. For civics, history, and law subjects where correct answers legitimately vary by country, this schema cannot represent the variation the deployment requires. The benchmark does provide country, subject, and level metadata fields per question [Q33], which would in principle allow country-stratified scoring — but this is a metadata affordance rather than a label-schema accommodation, and the metadata is null for some configurations [DATASET-D26, D27].

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
D8	— يوعدل رهوج يف مكحل رودص دعب ال عوفد وأ ضراع عازن يف ةرداصل وأ أديهمتمل مكحلأل فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D9	— عافدل اب ةفل كملم ةيموكحل ةطلسل او لدعل ربزو نم رارقب...نني كيردل ةيئاضق ال ةطرشل طباض ةفص لي وخت نكمي Moroccan judicial police powers — references "يكللمل كرذل" (Royal Gendarmerie of Morocco)
D10	— ةحيحص ةبوجل عيجم: مكحل رودص لبقي ندمل فرطل ل زانت اذا Moroccan civil party withdrawal in criminal proceedings
D26	All 5 examples: Country = null, Source = folderat.com, Subject = Islamic Studies — Islamic Studies "All" config examples have no country metadata — source appears to be generic Arabic quiz site
D27	All 5 examples: Country = null, Source = madinaharabic.com — Arabic Language General has no country metadata — sourced from online Arabic language learning site

Strengths

- Flat MCQ schema with 2–5 options [Q27] directly matches deployment's MCQ tutoring format
- Country, subject, and level metadata fields are present per question [Q33], enabling country-stratified analysis even without label-schema branching
- Subject-group taxonomy (STEM, Social Science, Humanities, Arabic Language, Other) [Q107] supports diagnostic per-subject reporting
- Education-level stratification (primary, middle, high, university) [Q102, Q103] aligns with the deployment's tier structure

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)

Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
Q102	"Table 8 presents the detailed zero-shot results across subjects and education levels, while Table 9, Table 10, Table 11 display the results with different prompts and alphabetic outputs (complementing the main result at Table 8)." (p.15)
Q103	"Zero-shot LLM performance (% accuracy) with English prompt and English alphabetic output, for each subject and education level." (p.15)
Q107	"Table 9 presents results organized by subject groups: STEM, Social Science, Humanities, Arabic Language, and Other." (p.16)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories (single correct option from 2–5) are regionally relevant for MCQ tutoring [Q27]; the format itself is appropriate. The categorical label space (A/B/C/D/E or Arabic equivalents [Q51, Q52]) is universally usable.

OO-2: Missing categories: no jurisdiction-specific correct-answer branching for law and civics [Q27], explicitly required by deployment [elicitation A2]; no madhab-specific branching for Islamic studies [gap_id 7]; no mechanism for 'correct in country X but not in country Y' [DATASET-D8, D34].

OO-3: The Driving Test configuration encodes country-specific traffic regulations as correct answers [DATASET-D35] — a positive example of country-tied ground truth, but applied inconsistently across other subjects where country variation also exists.

OO-4: Stakeholder-driven taxonomy redesign would benefit law and civics subsets to add jurisdiction tags. This is feasible given existing per-question country metadata [Q33] but not implemented in the published label schema.

OO-5: Documented taxonomy issues harming structural validity: (a) no jurisdiction tagging in label schema for law [DATASET-D34]; (b) Jordanian-default civics presented as universal [DATASET-D1, D2]; (c) no madhab labeling [gap_id 7]; (d) null country metadata for several configurations [DATASET-D26, D27].

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q33	"In the second stage, all workers were asked to manually scrape the data within a 2-month period. The task was to collect metadata, including the source (URL of the source document), country, subject, level, question, multiple-choice options, and the correct answer key." (p.4)
Q51	"In the case of English alphabetic output, we measure the probability of the first generated token being A, B, C, D, or E." (p.6)
Q52	"For Arabic, we measure the probability of the first generated token being @, H., h., X, or è." (p.6)
D1	What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics
D2	The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D8	— Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D26	All 5 examples: Country = null, Source = folderat.com, Subject = Islamic Studies — Islamic Studies "All" config examples have no country metadata — source appears to be generic Arabic quiz site

D27	All 5 examples: Country = null, Source = madinaharabic.com — Arabic Language General has no country metadata — sourced from online Arabic language learning site
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D35	رتم 50 دعب ىلع؟هعضو كيلىع نيأ، سكاعلا ريذحتلا ثلثم كيىلو كتبكرم ل طعت ل ا ح يف — UAE driving rule (50m warning triangle placement) — country-specific traffic regulation

Information Gaps

- Whether per-question country metadata is consistently populated across the full dataset (sample shows null for Islamic Studies All and Arabic Language General [DATASET-D26, D27])

Requires Expert Verification

- Whether subject-matter experts in law and civics deem the existing single-answer schema redeemable through metadata-stratified evaluation, or whether label-schema redesign is required

Remediation

Gap 1: Flat single-correct-answer schema [Q27] cannot represent jurisdiction-specific correctness, despite deployment requirement for country-tailored legal answers [elicitation A2].

Recommendation: Add a jurisdiction-tag overlay on top of ArabicMMLU's existing country metadata for law and civics subsets; rescore tutoring outputs against jurisdiction-conditioned ground truth rather than the flat label.

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)